

Operations

Player Turn: Move Spacecraft, any # of Free Actions, 1 Operation, in any order.

● Glitch Trigger

⚡ Felony

Operation	Summary	Details
Income	Get 1 Aqua	
Research Auction	Auction top card from any deck.	- Must have < 4 cards in Hand to start or to bid. - Auctioneer wins ties; pay auctioneer (Pool if auctioneer wins). - For each required support, take top card from that deck.
Free Market	Sell card from Hand or Black-Side card from LEO Stack.	- Get 3 Aqua for Hand card. - Get card's Exploitation Track value in Aqua for Black-Side card.
Boost	Play White-Side cards from Hand to LEO	- Pay Mass in Aqua. - Set radiator Orientation.
Site Refuel	● Add Fuel to Rocket, or FT to Stack.	- ISRU Refuel: Hydration – ISRU + 1 = water (<i>see separate table</i>). - Factory Refuel: 7 water; – Felony ⚡ at opponent's Factory. - Air-eater: Pac-Man 🐼 on Aerobrake: (5 - Fuel Consumption) water.
Prospect	● Evaluate and claim Site(s) (<i>see separate table</i>).	- ISRU ≤ Site Hydration (<i>see Prospecting Table</i>). - Roll 1d6: if ≤ Site Size, successful; or Site is Busted. - Raygun: prospect any/all Adjacent non-Atmospheric Sites. - Buggy: re-roll once OR prospect all Sites connected by buggy roads.
Industrialize	● Create a Factory.	- Decommission <i>any</i> robonaut and refinery at claimed Site. - Radiator supports not required and not Decommissioned.
ET Production	Produce Black-Side card from Hand to a Factory.	- Must match Spectral Type of Factory Site (C/S/M/V/D/H). - Set radiator Orientation.
Delivery	● Move Black-Side card from Factory to LEO	Pay FT from Factory = zones from Earth x2 (+1 if Site > 6 Size).

Prospecting

- 1) Check Site Hydration and robonaut ISRU on below table to see if prospecting is possible.

2) Roll less than or equal to Site Size on 1d6 to establish a Claim (Size > 5 = automatic success).

Prospect Operation					No Hydration
ISRU 4	Can prospect (1)	NO	NO	NO	NO
ISRU 3	Can prospect (2)	Can prospect (1)	NO	NO	NO
ISRU 2	Can prospect (3)	Can prospect (2)	Can prospect (1)	NO	NO
ISRU 1	Can prospect (4)	Can prospect (3)	Can prospect (2)	Can prospect (1)	NO
ISRU 0	Can prospect (5)	Can prospect (4)	Can prospect (3)	Can prospect (2)	Can prospect (1)

ISRU Refuel Operation: Refuel # of Water FT equal to number in parentheses above based on ISRU/Hydration.

Free Actions

● Glitch Trigger

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Free Action	Summary	Details
Cargo Transfer (Glitch Trigger when moving to 2 nd Stack) ●	Move cards and/or FTs between Co-located Stacks.	- Can form a new Rocket / Outpost Stack. - May Decommission (⚡ for Humans). - Refuel ∞ dirt FTs with any ISRU card at Factory or Site.
Internal Tankage	Convert between FTs and Fuel.	Can Decommission cards for dirt fuel.
Build Colony	Create permanent Colony at Factory.	Decommission Crew at Factory.
Claim Jump ⚡	Replace opponent's Claim with yours.	- You must have a Human present. - No opposing Factory or Human present.
Load Glory Chit	Load / Claim glory chit from Site previously unvisited by Humans.	- Humans must be present. - Each Crew can carry 1 glory chit at a time.
Voluntary Discard (Felony for Humans) ⚡	Discard cards and/or figures (1 Human per Turn max).	- Discard cards to bottom of patent decks. - Remove figures if their Stack is empty.
Glitch Repair	Remove Glitch token from Stack.	Human must be Colocated with Stack.
The Martian	Move Crew along buggy road.	Requires Operational buggy.

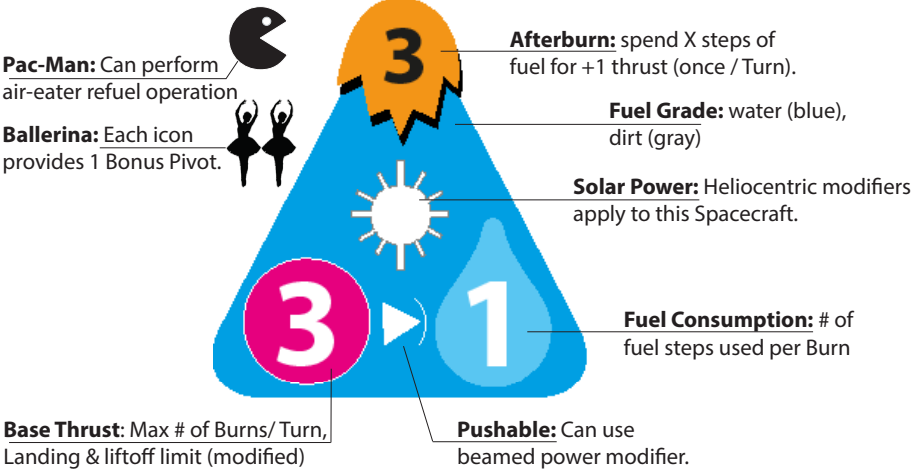
Thrust

Dry Mass: total Mass of all Rocket components/Cargo

Wet Mass: Dry Mass plus fuel (cannot be less than Dry Mass)

Fueling or adjusting Mass: use red arrows.

Consuming fuel: move fuel marker on black arrows.



Spacecraft Movement

● Glitch Trigger ⚡ Felony

Movement steps	Details
1. Activate an Operational thruster	Each Rocket can only use one thruster per Turn.
2. Starting net thrust = base thrust	Value in lower left corner of triangle.
3. Modify for afterburn	If thruster has afterburn at top, may spend cost for +1 net thrust (once/turn). (Afterburning provides 1 Therm of cooling for the thruster support chain.)
4. Modify for Wet Mass	+2 to -2 based on weight class (Wet Mass after adjusting for afterburn).
5. Modify for Heliocentric Zone	Only if thruster or supports have solar icon ☀.
6. Modify for beamed power ▶	+1 if thruster has push icon and player has beamed power / Powersat.
7. Modify for supports 🔧	Net thrust, fuel consumption, and/or solar icon ☀ can be modified. Ignore movement-modifiers for supports needed for radiators and/or reactors. Solar-Power supports ☀ can never be used in the Neptune Heliocentric Zone without Powersat.
8. Result is net thrust	Net thrust will not change even if Rocket composition changes mid-move (e.g. Cargo is destroyed).
9. Note Bonus Pivots ⚡	Pivots which can be performed without requiring Burns.
10. Note fuel consumption	Fuel consumption value in lower right corner of triangle.
11. Move	Move through Pivots and Burns until # of Burns (or fuel) are used up; then Coast as far as desired (including Landing).

Each Burn expends a # of fuel steps equal to the fuel consumption; Net thrust = max # of Burns per Turn.

Solar System Map

● Glitch Trigger ⚡ Felony

Name	Icon	Restrictions
Hohmann Intersections (no circle)		• Freely move straight through (if possible) or stop. • Must Pivot (2 Burns) to change direction. • If begin Turn on Hohmann Intersection, free to move in any direction.
Lagrange Point Intersection (circle)		• Freely move straight through, stop, or change direction.
Lagrange Burn		• 1 Burn to enter.
Lander/ Half Lander Burn		• 1 Burn to enter Lander. • 1 Burn (½ fuel consumption) to enter Half Lander. • Cannot end move on these Spaces.
Site (Land and Liftoff) (⚡ using opponent's Factory)		Free to land or liftoff if net thrust > Site Size with the following exceptions: 1. Aerobrake path landing: Hazard Roll required. 2. Factory-assist landing or liftoff: Hazard Roll required (unless you have Powersat) and cannot move to/from lander burns. 3. Rocketplane liftoff (H6c), can use factory-assist as above for Atmospheric liftoff (pay 2 * Wet Mass in FTs).
Synodic Comet Site (Land and Liftoff)		As above for Sites; additionally, cannot enter or exit unless Sunspot Cube is in matching-colored season.
Flyby ●		Gain Bonus Burns which don't use fuel (can't be used for lander burns).

Starting Advice

What Not To Do	What To Do Instead
Go to a large Site with no way to liftoff.	• Bring a stronger thruster (net thrust larger than Site Size). • Go to a smaller Site.
Go to a series of Size 1 Sites, hoping to roll a  while prospecting. <i>(This incudes comets, which also require a matching season to enter/exit.)</i>	• Bring a raygun robonaut to a cluster of adjacent Sites. • Go to a larger Site.
Go to a smaller Site (Size 5 or less) with no backup plan if the Size roll fails.	• Bring enough fuel to get home or to another Site. • Bring a buggy robonaut to improve the chances on the Size roll. • Bring a robonaut that can ISRU refuel on the Site.
Go to Venus without an ISRU 0 robonaut. <i>(Venus has zero site hydration.)</i>	• Go to Mars (any robonaut will work there).
Go to Mercury without a very efficient thruster. <i>(Mercury is farther than it looks, energy-wise.)</i>	• Consider going to Vesta instead.

Suggested Early Destinations

Spectral Type	ISRU	Destination	Size	Path	Burns / Hazards	Recommended
C	4	Ceres	6	Blue path.	4.5	ISRU 3 Raygun.
	4	Mars	10	Red path.	3, 1 Hazard, 1 Aerobrake	ISRU 3 Buggy.
S	1	Flora	4	Purple path.	3	Buggy or Raygun.
	1	Luna	9	HEO to Luna.	2	Buggy or Raygun; 1st player's permission or Felony.
	3	Karin cluster	1	Purple path.	3, 1 Hazard	ISRU 2 Raygun.
M	3	Hertha	3	Purple path.	3	Buggy.
	3*	Lutetia	3*	Yellow path.	4	Buggy. <i>Check map, may be Size 4 1 hydration!</i>
V	2	Vesta	6	Blue path.	4.5	ISRU 1 Raygun.
	4	Callisto	8	Purple path to Sol-Mars L4, Teal path, Jupiter flyby.	5, 1 Hazard, 2 Aerobrakes.	Buggy.
D	3	Hektor	3	Purple path.	5, 1 Hazard.	ISRU 3 Raygun or Buggy.
	4	Echeclus	2	Purple path to Sol-Mars L4, Teal path, Jupiter flyby.	4, 2 Hazards, 2 Aerobrakes.	Buggy.
H	0	Luna	9	HEO to Luna.	2	Buggy or Raygun; 1st player's permission or Felony.

Signpost Summary

Destination	Color	Burns	Landing / Liftoff / Net Thrust Required	Hazards	What's There?
Mercury	Orange	7	11		10S (0 hydration) / 10V (3 hydration), Push Factory.
Venus	Indigo	3	Aerobrake / 12 / Acetyline rocket liftoff	Aerobrakes, Hazard.	11C (0 hydration), Astrobiology, Push Factory.
Mars	Red	3	Aerobrake / 11	Aerobrake, Hazard.	10C (3 hydration) / 10C (4 hydration).
Hertha	Purple	3	4		3M (3 hydration).
Achilles	Purple	5	3		2D (3 hydration).
Callisto	Purple	8	9		8C (4 hydration), Subsurface Ocean / 8V (4 hydration).
Ceres	Teal	4.5	7		6C (4 hydration), Subsurface Ocean.
Lutetia	Yellow	4	4		3M (3 hydration) <i>(may be 4M 1 hydration)</i>
Aeneas	Green	6	3		2D (3 hydration)
Enceladus	Green	7	6	Aerobrakes, Hazards.	5M (4 hydration), Subsurface Ocean.

Game Setup and Turn Structure

Game Duration: 4 Seniority Disks

Phase	Description
Player Turn Phase (D1)	Each player takes their Turn. In any order: spacecraft movement, 1 Operation, free actions.
Advance Sunspot Cycle Phase (D2)	Advance cube on Sunspot Cycle
	If the cube crosses an "Event" Threshold, roll for an Event (see below)
	If the cube crosses the "Seniority" Threshold, remove a Seniority Disk. (If all disks removed, the game ends.)

Threats

Threat	Triggers	FINAO	Roll Effect
Hazard	Aerobrake Hazard Spaces; Crash Hazard Spaces; Factory-Assist Landing/Liftoff	Yes	1d6: Decommission Stack on a  .
Glitch 	Flybys; Cargo Transfer; Site Refuel; Prospect; Industrialize	No	1d6: Decommission collocated cards with matching rad-hardness; roll before triggering event occurs.
Belt	Radiation Belt Spaces 	No	1d6, subtract net thrust, +2 if in Sunspot Cycle in red: Decommission cards with lower rad-hardness
CME	Solar Flare Event (roll of  or  during red Sunspot Cycle)	No	1d6 + Heliocentric Zone modifier: Decommission cards with lower rad-hardness (LEO, Cyclor, non-Sol Belt Spaces, Sites unaffected).

FINAO = Failure Is Not An Option; spend 4 Aqua before roll to ignore Hazard.

Events

Roll	Event	Effect
 or 	Inspiration	Put top card of each deck on bottom of that deck.
	Glitch	Each player places Glitch token on glitch-free non-LEO Stack with most cards.
	Pad Explosion / Space Debris	Each player Decommissions heaviest card in LEO. (Crew, Black-Side cards are immune.)
 or 	Anarchy (Blue)	• All Faction privileges lost, all players have Felonious for rest of blue season.
	Budget Cuts (Yellow)	Each player Discards a card from Hand.
	Solar Flare (Red)	Make a global CME Roll (see CME Threat above).

Felonies

Felony	Event	Defense
Claim Jump (Free Action)	Replace opponent's Claim disk with yours or claim Luna.	Opposing Humans present
Factory Hijack	Refuel, ET Produce, factory-assist landing/liftoff using opponent's Factory.	Opposing Humans present
Murder / Suicide	Decommission your Humans (once / Turn).	N/A

You must have a Human present to commit any Felony.

Game Scoring

Scoring	Points (tiebreaker: most Aqua)
Token VP	1 VP for each wooden/plastic Token (Rocket, Claim disk, Factory cube, Colony dome) on map.
Bonus VP	Factory: +8 / +5 / +4 VP each, according to Exploitation Track
	Colonies: +1 VP on Astrobiology; +2 VP on Submarine.
	Glory chits: VP per chit based on which side is faceup.